

APS110 – Lecture 12

See hand written notes for drawings.

Connection between structure and properties

- Iron is BCC
- Is Young's modulus higher along cube diagonals or edges of the cube?
- Recall: Young's modulus is equivalent to the first derivative of the force-separation curve at $r=r_0$
- The thing we know that is special about the cube diagonal, is that the atoms are making contact with one another → close-packed direction
 - Spacing between atoms is $r=r_0$ (atoms are touching)
- Edges → equilibrium spacing is not close-packed
 - The distance between atoms, r , is greater than r_0
 - If we take the derivative, it would be to the right (slope will be less)
- For Iron:
 - $E_{\text{cube diagonal}} = 273 \text{ GPa}$
 - $E_{\text{edge}} = 125 \text{ GPa}$
 - You can see the modulus along the edges is less than half!

Microstructure

- Most of the metals we deal with are polycrystalline
- Properties of the polycrystalline metal are **an average** of all grain orientations
 - When crystals form, they grow from crystal nuclei and grow outward, until they meet other crystals
 - These meeting points are grain boundaries
 - As such, the orientation of these crystals are random between grains and so the overall structure of the metal is an average of all the different crystal orientations of the different grains

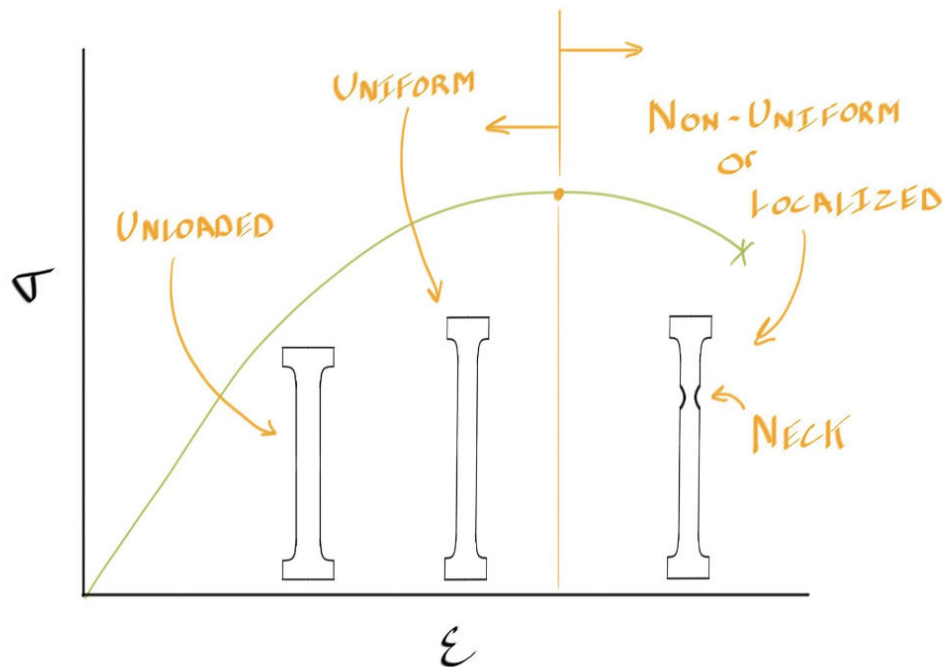
How does plastic deformation occur?

- We have a close packed plane, and when force is applied the planes slide past each other
 - The end result is the plastically deformed planes
- Why are ceramics brittle?
 - When a force is applied, to deform the structure like the above example (planes sliding past each other), the ceramic would have to go through this high energy intermediate, where there is significant **charge repulsion**
 - So, ceramics are very difficult to plastically deform → at room temp, fracture occurs before plastic deformation

Stress-strain curve

- The linear elastic region has a slope of E

- There is a peak to the curve, which is a moment in the stress state of the sample where **necking** occurs
 - Tensile test – as you stretch the sample, initially you'll need a lot of force applied
 - At a certain point, something happens and the force required to continue stretching is significantly reduced → necking
 - You get this localized plastic deformation somewhere along the reduced section of the tensile specimen
 - The peak of the curve is the highest the stress is going to get, we call this the **Ultimate Tensile Strength (UTS)**
 - At the stress level of this peak, you get σ_{UTS}



- σ_f = stress at which fracture occurs, "**fracture strength**"
- How to define plastic deformation?
 - On the stress-strain curve, how do we describe the delineation between where linearity stops?
 - The common standard that we use for most metals is the **0.2% offset yield strength**